

Node.js Built-in Modules Cheat Sheet

By Eric H

Node.js comes with many built-in modules for file system, networking, utilities, and more.

They provide essential building blocks for backend development.

This cheat sheet gives you a quick reference to the most commonly used Node.js core modules.

Tip

Production-ready SaaS boilerplate  <https://www.thedevspace.io/boilerplate>

1. `fs` (File System)

Read/write files and directories.

API Overview

- `fs.readFile(path, [options], callback)` – Read file asynchronously.
- `fs.writeFile(path, data, [options], callback)` – Write file asynchronously.
- `fs.readdir(path, callback)` – List directory contents.
- `fs.mkdir(path, [options], callback)` – Create directory.

Examples

```
1  const fs = require("fs");
2
3  // Read a file
4  fs.readFile("file.txt", "utf8", (err, data) => {
5    if (err) throw err;
6    console.log(data); // Output file contents
7  });
8
9  // Write a file
10 fs.writeFile("output.txt", "Hello, Node!", (err)
    => {
11   if (err) throw err;
12 });
```

2. `path`

Utilities for file and directory paths.

API Overview

- `path.join(...paths)` – Join paths.
- `path.resolve(...paths)` – Resolve to absolute path.
- `path.basename(path)` – Get file name.
- `path.dirname(path)` – Get directory name.

Examples

```
1 const path = require("path");  
2  
3 const fullPath = path.join(__dirname, "file.txt");  
4 console.log(fullPath);
```

3. `os`

System info and utilities.

API Overview

- `os.platform()` – OS platform.
- `os.cpus()` – CPU info.
- `os.totalmem()` – Total memory.
- `os.homedir()` – Home directory.

Examples

```
1  const os = require("os");
2
3  console.log(os.platform()); // e.g. darwin
4  console.log(os.homedir()); // e.g. /Users/xxx
```

4. http

Create HTTP servers and clients.

API Overview

- `http.createServer([requestListener])` – Create HTTP server.

Examples

```
1  const http = require("http");
2
3  const server = http.createServer((req, res) => {
4    res.end("Hello, world!");
5  });
6  server.listen(3000);
```

5. `events`

Event emitter for custom events.

API Overview

- `EventEmitter` class: `.on()`, `.emit()`, `.off()`

Examples

```
1  const EventEmitter = require("events");
2  const emitter = new EventEmitter();
3
4  emitter.on("greet", (name) => {
5    console.log(`Hello, ${name}!`); // Hello, Alice!
6  });
7
8  emitter.emit("greet", "Alice");
```

6. `crypto`

Cryptographic utilities.

API Overview

- `crypto.randomBytes(size, callback)` – Generate random bytes.
- `crypto.createHash(algorithm)` – Create hash object.

Examples

```
1 const crypto = require("crypto");
2
3 const hash =
  crypto.createHash("sha256").update("password").digest("hex");
4 console.log(hash); // e.g.
  5e884898da28047151d0e56f8dc6292773603d0d6aabbdd62a1
  1ef721d1542d8
```